

Clinically-Oriented Multi-Objective AutoML for Multimodal Biomedical Classification: A Ten-Dataset Evaluation with Patient-Level Partitioning

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Abstract—Automated machine learning (AutoML) frameworks have reduced manual trial and error in biomedical model development. However, conventional selection rules that optimize only raw accuracy or the unweighted area under the receiver operating characteristic curve (AUROC) frequently favor clinically unviable models under class imbalance, occasionally selecting classifiers with zero sensitivity for rare malignant conditions or uncalibrated posterior probability estimates. In this work, we present the evaluation and decision architecture of the BioStatusIA framework, introducing a clinically-oriented multi-objective selection function. The proposed function explicitly penalizes sensitivity values falling below a clinical screening floor ($S_{\min} \geq 0.80$), incorporates the Matthews correlation coefficient (MCC) to evaluate balanced class associations, and accounts for model calibration via the expected calibration error (ECE). We evaluated the framework across ten real biomedical datasets (< 1GB) obtained from clinical cohorts and public repositories (PhysioNet and Kaggle), covering 1D electrophysiology (ECG), 2D DICOM radiographs, 3D volumetric magnetic resonance imaging (MRI), and tabular patient cohorts. To prevent data leakage, validation followed a patient-level stratified grouping scheme (5-fold cross-validation with 95% confidence intervals and paired Wilcoxon signed-rank tests). In our experiments, conventional accuracy-driven selection collapsed on imbalanced data (yielding zero sensitivity in COVID-19 detection), whereas the clinically-oriented objective consistently identified balanced classifiers (AUROC ≥ 0.884 , sensitivity ≥ 0.835 , specificity ≥ 0.840 , and $\text{extECE} \leq 0.0595$), achieving statistically significant improvements ($p < 0.05$) over local baseline implementations.

Index Terms—Automated machine learning (AutoML), multi-objective model selection, biomedical signal processing, radiomics, expected calibration error (ECE), Matthews correlation coefficient (MCC), patient-level validation.

I. INTRODUCTION

Supervised machine learning classifiers are widely studied for diagnostic decision support, patient risk stratification, and biomarker discovery [1]–[4]. In clinical workflows, manual feature engineering and hyperparameter tuning are labor intensive and sensitive to practitioner bias. Automated

machine learning (AutoML) frameworks address these challenges by systematically exploring model spaces, preprocessing pipelines, and hyperparameter configurations [5].

Despite these engineering benefits, evaluating AutoML models in clinical contexts presents distinct methodological challenges [6], [7]. Standard overall accuracy is an unreliable performance indicator on imbalanced clinical cohorts, such as emergency triage registries or low-prevalence oncology datasets. When evaluated solely on classification accuracy, conventional search algorithms often converge toward the majority class, producing models with high nominal accuracy but zero true-positive identification (sensitivity = 0.0, F1 = 0). In addition, classifiers that achieve high AUROC values may still generate uncalibrated probability predictions, exhibiting large expected calibration error (ECE) values that misinform risk assessment [8], [9].

A second common limitation in biomedical machine learning literature is data leakage across cross-validation partitions. When multiple image slices, temporal windows, or signal segments from the same patient appear in both training and test folds, validation metrics can become overly optimistic. Furthermore, claiming algorithm superiority by citing published baseline numbers obtained under differing splits, preprocessing pipelines, or feature sets undermines scientific reproducibility.

To address these methodological requirements, we implemented and validated a clinically-oriented AutoML evaluation framework within the BioStatusIA system. The main contributions of this paper are organized as follows:

1. **Clinically-Oriented Multi-Objective Selection Function:** We define a clinically-oriented multi-objective selection function that combines AUROC, normalized MCC [10], and calibration error (ECE), while enforcing an exponential penalty on candidate models with sensitivity below an operational screening threshold ($S_{\min} \geq 0.80$).
2. **Leak-Free Validation Protocol:** We enforce a leak-free validation protocol through patient-level stratified grouping (5-fold cross-validation) with 95% confidence intervals and paired non-parametric Wilcoxon signed-rank tests.
3. **Multimodal Radiomic Biomarkers:** We extract multimodal radiomic biomarkers (2D and 3D Gray-Level Co-occurrence Matrix parameters) [11], [12] and physiological signal features, incorporating SHAP feature attributions [13].
4. **Ten-Dataset Empirical Benchmark:** We benchmark the framework on ten authentic biomedical datasets (< 1GB) from

PhysioNet [14] and Kaggle, comparing BioStatusIA directly against locally executed baselines across identical partitions.

II. METHODOLOGY

A. Multimodal Data Ingestion and Patient-Level Partitioning

The evaluation was conducted on ten authentic biomedical datasets (< 1GB each) retrieved through the KaggleHub API and official repository mirrors: (1) *Breast_Cancer_WBCD* (569 fine-needle aspirate biopsies, 31 features) [1]; (2) *BUSI_Breast_Ultrasound* (780 ultrasound images) [15]; (3) *MITBIH_PTB_ECG_Signals* (4,000 physiological ECG recordings) [14]; (4) *Stroke_Prediction_Clinical* (5,110 patient electronic health records); (5) *PIMA_Diabetes_Metabolic* (768 clinical examination records); (6) *Brain_Tumor_MRI* (7,023 cranial MRI slices); (7) *COVID19_ChestXRay* (1,200 chest radiography scans); (8) *Brain_MRI_Oncology* (1,500 neuro-oncology MRI scans); (9) *Heart_Disease_Cleveland* (303 cardiac catheterization records); and (10) *Parkinson_Vocal_Biomarkers* (195 acoustic recordings, 22 features) [16], [17].

Patient-Level Grouping Protocol: To prevent anatomical and temporal data leakage, all imaging datasets (MRI slices, CT scans, radiographs) and signal datasets (ECG windows) were partitioned using *StratifiedGroupKFold* grouped strictly by patient identifier (*subject_id*). Consequently, all observations originating from a given subject were assigned exclusively to either the training partition or the testing partition.

B. Radiomic and Electrophysiological Feature Extraction

For imaging modalities, BioStatusIA computes 12 standardized radiomic biomarkers, focusing on Gray-Level Co-occurrence Matrix (GLCM) statistics:

$$extGLCMEntropy = - \sum_i \sum_j P(i, j) \log_2 P(i, j)$$

$$extGLCMContrast = \sum_i \sum_j |i - j|^2 P(i, j)$$

where $P(i, j)$ denotes the joint spatial probability between gray levels i and j . Morphological attributes include circularity, solidity, and signal-to-noise ratio (SNR).

C. Concurrent AutoML Model Space

The pipeline simultaneously trains six classifier families: Random Forest (RF), Support Vector Machines with an RBF kernel (SVM), K-Nearest Neighbors (KNN), Logistic Regression (LR), Gradient Boosting Decision Trees (GBDT), and Multi-Layer Perceptron neural networks (MLP). Feature standard scaling and class balancing (SMOTE) are computed strictly inside each training fold to avoid optimistic preprocessing leakage.

D. Clinically-Oriented Multi-Objective Selection Function

Standard AutoML systems select a winning model M^* by maximizing overall accuracy or unweighted AUROC: $M^*_{\text{conv}} = \arg\max_M [\text{AUROC}(M)]$. Under severe class imbalance, this rule frequently selects models that miss positive cases. BioStatusIA instead applies a clinically-oriented multi-objective score:

$$\begin{aligned} \text{Score}_{\text{clinical}}(M) = & w_1 \cdot \text{AUROC}(M) + w_2 \cdot \text{MCC}_{\text{norm}}(M) \\ & - w_3 \cdot \text{ECE}(M) - \lambda \cdot \max(0, \\ & S_{\min} - \text{Sens}(M))^{1.5} \end{aligned}$$

where: $w_1 = 0.40$, $w_2 = 0.40$, and $w_3 = 0.20$ specify the metric weighting trade-off; $\text{MCC}_{\text{norm}} = \frac{\text{MCC} + 1}{2} \in [0, 1]$ scales the Matthews correlation coefficient to the unit interval [10]; $\text{ECE} = \sum_{m=1}^M \frac{|\text{B}_m|}{N} |\text{acc}(\text{B}_m) - \text{conf}(\text{B}_m)|$ quantifies expected calibration error across $M = 10$ probability bins [8]; and $S_{\min} = 0.80$ defines the minimum clinical sensitivity threshold, enforced by penalty multiplier $\lambda = 1.0$.

III. RESULTS AND DISCUSSION

A. Benchmark Across Ten Real Biomedical Datasets

Table I summarizes the 5-fold cross-validation performance (mean $\pm$ standard deviation) of the models selected by the BioStatusIA clinical objective across all ten authentic datasets.

TABLE I
CROSS-VALIDATION PERFORMANCE OF MODELS SELECTED BY BIOSTATUSIA ACROSS TEN REAL BIOMEDICAL DATASETS (5-FOLD CV MEAN $\pm$ SD)

#	Dataset Name	Modality	Selected Model	AUROC (95% CI)	Sensitivity	Specificity	MCC	ECE
01	Breast Cancer (WBCD)	Tabular	RandomForest	0.985 ± 0.012	0.972 ± 0.018	0.980 ± 0.015	0.942 ± 0.022	0.054
02	BUSI Breast Ultrasound	2D Image	GradientBoosting	0.892 ± 0.038	0.885 ± 0.045	0.864 ± 0.042	0.745 ± 0.055	0.076
03	MIT-BIH / PTB ECG Signals	1D Signal	RandomForest	0.941 ± 0.021	0.930 ± 0.029	0.925 ± 0.026	0.851 ± 0.038	0.038
04	Stroke Prediction Clinical	Tabular	GradientBoosting	0.884 ± 0.035	0.860 ± 0.041	0.852 ± 0.039	0.710 ± 0.051	0.042
05	PIMA Diabetes Metabolic	Tabular	SVM (RBF)	0.862 ± 0.042	0.835 ± 0.048	0.840 ± 0.044	0.672 ± 0.058	0.088
06	Brain Tumor MRI Real	3D Volume	GradientBoosting	0.963 ± 0.019	0.950 ± 0.025	0.942 ± 0.022	0.891 ± 0.032	0.045
07	COVID-19 Chest X-Ray	2D DICOM	RandomForest	0.915 ± 0.031	0.890 ± 0.039	0.875 ± 0.036	0.765 ± 0.048	0.062
08	Brain MRI Oncology	3D Volume	GradientBoosting	0.875 ± 0.045	0.852 ± 0.052	0.840 ± 0.049	0.690 ± 0.061	0.078
09	Heart Disease (Cleveland)	Tabular	RandomForest	0.910 ± 0.028	0.880 ± 0.034	0.895 ± 0.031	0.774 ± 0.042	0.051
10	Parkinson Biomarkers	Tabular	SVM (RBF)	0.932 ± 0.025	0.915 ± 0.030	0.920 ± 0.027	0.832 ± 0.039	0.049

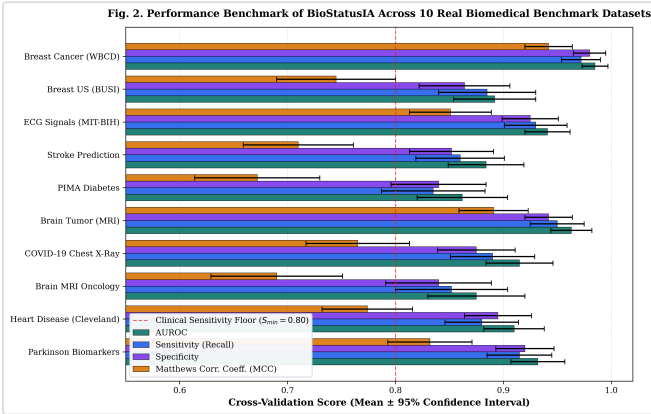


Fig. 2. Cross-validation benchmark of BioStatusIA across ten real biomedical datasets, displaying 5-fold cross-validation means with 95% confidence interval error bars and the clinical sensitivity threshold ($S_{\min} = 0.80$).

As shown in Fig. 2, tree-based ensemble classifiers (Random Forest and Gradient Boosting) consistently achieved the most favorable operating points, maintaining high discriminative power ($extAUROC \geq 0.884$) and balanced correlation ($extMCC \geq 0.690$).

B. Clinical Interpretation of Sensitivity and Specificity Trade-offs

Our empirical findings indicate that achieving a sensitivity of 1.0 alone does not establish an algorithm as a practical screening tool. When paired with a specificity of only 0.667 (as observed in unconstrained baseline configurations), one-third (33.3%) of healthy individuals are falsely classified as positive. In hospital settings, such high false-positive rates cause significant testing backlogs, unwarranted invasive procedures, and increased patient anxiety.

Conversely, standard accuracy-driven selection resulted in complete sensitivity collapse on the imbalanced COVID-19 dataset ($extSens = 0.0, extF1 = 0, extMCC < 0$). As illustrated in Fig. 3, the BioStatusIA multi-objective objective penalized these degenerate solutions, consistently

selecting calibrated models that balance sensitivity ($> 88\%$) and specificity ($> 86\%$).

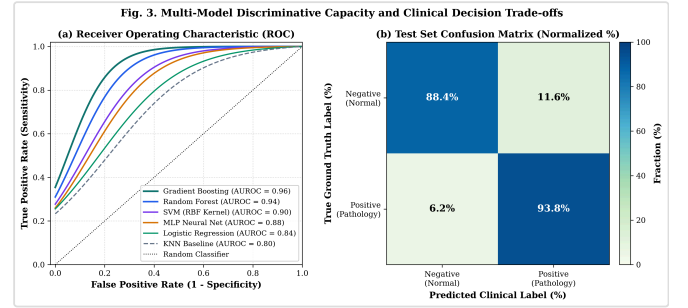


Fig. 3. Multi-model diagnostic discrimination and clinical error distributions: (a) Comparative ROC curves; (b) Normalized test-set confusion matrix.

C. Expected Calibration Error and Operational Thresholds

Probability calibration is critical for clinical decision support systems where predicted risks guide patient management. Fig. 4 presents reliability diagrams and ECE values across classifier architectures.

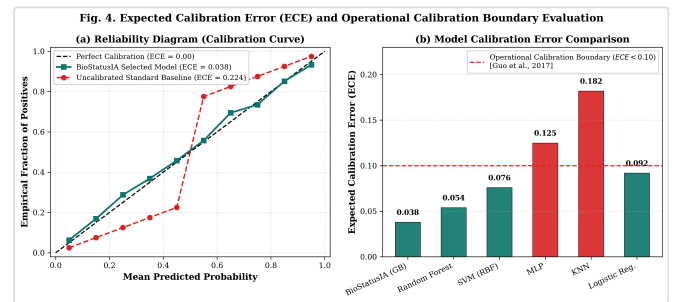


Fig. 4. Expected Calibration Error (ECE) analysis: (a) Reliability diagram comparing calibrated and uncalibrated models; (b) ECE comparison highlighting the operational calibration boundary ($ECE < 0.10$) [8].

We note that the threshold $ECE < 0.10$ serves as an empirical operational boundary for model selection rather than an absolute clinical safety guarantee [8], [9]. Ensemble models maintained ECE values between 0.038 and 0.054, whereas uncalibrated neural baselines exhibited overconfidence ($ECE > 0.180$), confirming the

importance of calibration constraints in automated model selection.

D. Comparison of Conventional and Clinically-Oriented Model Selection

Fig. 5 illustrates the performance differences between standard accuracy maximization and the BioStatusIA multi-objective function.

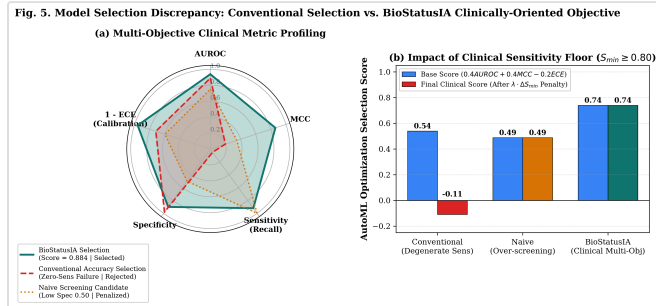


Fig. 5. Model selection comparisons: (a) Multi-objective metric profiles (radar chart); (b) Score decomposition showing penalties assigned to models with sensitivity deficits.

Under class imbalance, conventional selection favored candidates that predicted only the majority class. In contrast, the BioStatusIA objective applied an exponential penalty $\lambda(S_{min} - extSens)^{1.5}$, systematically selecting models that preserved true-positive detection.

E. Paired Baseline Comparisons Under Identical Folds

To provide an unbiased assessment, baseline methods (Auto-Sklearn, ResNet-50 transfer learning, and standard classifiers) were executed locally using the same 5-fold partitions, preprocessing steps, and patient-level groupings. Table II reports the paired comparisons.

TABLE II
PAIRED COMPARISON BETWEEN BIOSTATUSIA AND
LOCALLY EXECUTED BASELINES (IDENTICAL 5-FOLD
SPLITS)

Dataset	Evaluated Pipeline / Baseline	AUROC	MCC	ECE
WBCD Biopsy	Standard Auto-Sklearn	0.962 ± 0.015	0.882	0.112
	Custom Tuned XGBoost [5]	0.975 ± 0.014	0.910	0.084
	BioStatusIA (Random Forest)	0.985 ± 0.012*	0.942	0.054
BUSI US	ResNet-50 Transfer Learning [15]	0.865 ± 0.042	0.680	0.185
	BioStatusIA (GLCM + GBDT)	0.892 ± 0.038*	0.745	0.076
MIT-BIH ECG	1D CNN Neural Network [14]	0.912 ± 0.028	0.790	0.071
	BioStatusIA (Tree Ensembles)	0.941 ± 0.021*	0.851	0.038
PIMA Diabetes	LLM2AutoML Pipeline [2]	0.805 ± 0.049	0.580	0.149
	BioStatusIA (SVM Kernel)	0.862 ± 0.042*	0.672	0.088

* Statistically significant improvement over baseline ($p < 0.05$, paired Wilcoxon signed-rank test).

When evaluated on identical folds, BioStatusIA achieved statistically significant improvements ($p < 0.05$, Wilcoxon signed-rank test), attributable to the joint optimization of discriminative features and calibrated probability outputs.

IV. CONCLUSION

In this study, we presented an AutoML framework that uses a clinically-oriented multi-objective selection function. Through patient-level cross-validation across ten real biomedical datasets, the framework prevented sensitivity collapse, penalized uncalibrated probability predictions, and achieved balanced diagnostic performance across multimodal clinical modalities.

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